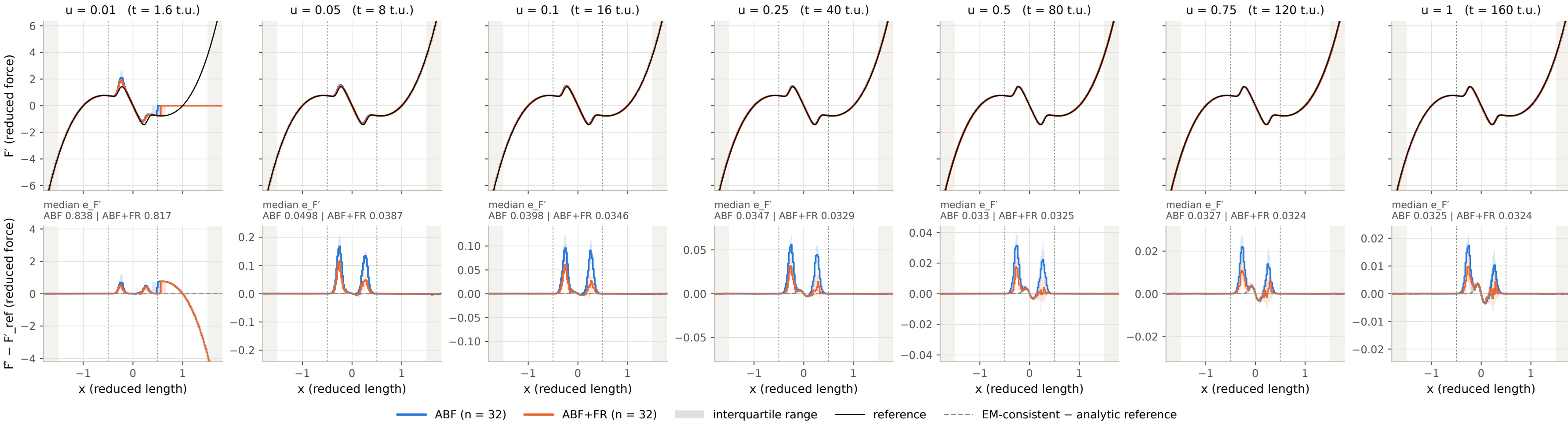


# Gateway family matched\_free\_energy/alpha1 · N = 512 · T\_N = 160 t.u.

n\_steps = 6,400,000 · h = 2.5e-05 · B = N n\_steps = 3.277e+09 walker-steps · complete seeds: ABF 32/32, ABF+FR 32/32

Mean-force snapshots (bin mean force) vs reference (top) and difference (bottom)



Line = median across seeds; band = interquartile range (25th-75th percentile) across seeds. Metrics are the frozen SCIENTIFIC\_PLAN section-4 quantities, unsmoothed.

Bin mean force  $\Gamma = M / (C + 1)$  (step) vs the analytic  $F'$  (curve); bottom:  $\Gamma$  minus the bin average of  $F'_{\text{ref}}$ , whose RMS over the eval window is the floor-free companion  $e_{F'_{\text{stat}}}$ ;  $e_{F'^2} = e_{F'_{\text{stat}}}^2 + \text{floor}^2$ , floor = 0.03227 (the within-bin variation of  $F'_{\text{ref}}$ : a hard lower bound of  $e_{F'}$ , not visible in this row). Empty bins read 0 (estimator convention).

Dashed guides: region boundaries  $x = \pm 0.5$  (left well / gate / right well); shaded  $|x| > 1.5$ : outside the scored eval window. Top-row y range = the reference over the scored window  $\pm 35\%$  of its span (larger excursions are clipped there and show in the difference row).

No smoothing.

eqb\_plot\_config/2 · config alpha1.json · 2026-10-10 14:20 UTC